

Report of Industrial Training

at

Indian Institute of Science, Bangalore

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

Submitted by

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200907282



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MANIPAL INSTITUTE OF TECHNOLOGY

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MANIPAL – 576104, KARNATAKA, INDIA

August 2023



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4th Aug 2023

To Whom It May Concern

This is to certify that **Mr. Ayush Shukla**, 4th year BTech student in Electronics and Communication at Manipal Institute of Technology, Manipal, India did his summer internship in the Department of ECE, Indian Institute of Science, Bangalore from 1st June to 31st July 2023. He was working under my supervision in the Numerics of Integrated Circuits and Electromagnetics (NICE) laboratory.

Ayush worked on a project titled "multi-dimensional visualization and optimization for Electromagnetic Compatibility analysis".

Thanking you,
Sincerely

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ACKNOWLEDGMENT

I would like to express my gratitude and sincere thanks to **Dr. Dipanjan Gope (Associate Professor, ECE, IISC)** and **Dr. Jan Hansen (Visiting Professor, TU Graz)** for their guidance and support during my course of internship at the Department of Electrical and Communication Engineering, Indian Institute of Science, Bangalore

During the internship, I was working with a team consisting of Research Scholars from the Indian Institute of Science and Technical University of Graz, Austria. I had an excellent exposure on Electromagnetic Compatibility and Electromagnetic Interference and on further on different mathematical techniques such as Surrogate modelling and Uncertainty quantification.

My sincere thanks are also to **Mr. Akash, Mr. Samipendu Das** and **Mrs. Suguna [PhD scholars, IISC]** for providing me with all the support and guidance whenever and wherever required during the period of internship.

I am also grateful to Silicon Austria Labs (SAL) as this work has been supported by the “University SAL Labs”, an initiative of Silicon Austria Labs and its Austrian partner universities for applied fundamental research for electronic based systems.

I am grateful to the ECE Department, MIT Manipal for granting me the chance to undertake this internship.

Lastly, I extend my thanks to my parents, well-wishers, and friends for their unwavering support and assistance throughout this journey.

Ayush Shukla

Introduction

DC-DC converters are an integral part of any electronic system for power distribution and are the primary source of electromagnetic interference (EMI). The electronic system design for electromagnetic compatibility involves using an EMI filter with a DC-DC converter to reduce the conducted emissions.

Analysis of filter performance at the design stage is crucial to achieve an optimum filter in terms of volume and performance. However, the parasitics of components, printed circuit board (PCB), and mutual coupling between the components alter the actual performance of the EMI filter in the manufactured product. To this effect, different methodologies based on equivalent circuit models, electromagnetic models, and artificial neural networks are proposed to model EMI filters in the literature.

Modeling using equivalent circuit models / 3D modeling methods is time-consuming because of the broad frequency range of operation of EMI filters.

The surrogate model / meta model approach has gained importance to avoid excessive computation times by fast mathematical approximations such as kriging or polynomial chaos expansion.

In a so-called wideband kriging (WBK) based surrogate model is used to carry out the optimization of a common mode choke. We also extended the WBK to estimate Common Mode (CM) emissions of a power system combined with EMI filter.

Owing to multiple ways of obtaining the surrogate models, the work I carried out compares different surrogate modeling approaches for estimating EMI filter performance using UQLAB uncertainty Quantification framework.

Training Details

Title:

Comparison of Surrogate Modeling Approaches for Estimation of EMI Filter Insertion Loss

Abstract and Contribution:

The growing use of silicon carbide and gallium nitride transistors is leading to a reduction in the size of power electronic systems due to their faster switching and clock times.

However, this progress is accompanied by increased electromagnetic emission, which necessitates additional expenses for designing EMI filters. Achieving an optimal solution that satisfies both requirements demands significant computational resources.

To expedite this process, metamodeling techniques are employed to approximate complex physical models with simpler mathematical equivalents derived from training data. In instances where the physical models are computationally demanding, it becomes crucial to identify the most efficient methods.

We have compared various surrogate modelling techniques, such as Kriging, Polynomial Chaos Expansion, Polynomial Chaos Kriging, and Wideband Kriging. Our findings suggest that while Wideband Kriging offers the highest level of accuracy, it also involves the longest model generation time.

Key words:

EMI/EMC, Surrogate model, Kriging, Wideband Kriging, Silicon Carbide, Gallium Nitride, Meta model, EMI filter, Polynomial Chaos, and Polynomial Chaos Kriging

EMI/EMC

Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) are critical concepts in the field of electrical engineering that pertain to the management and control of electromagnetic radiation and its impact on electronic devices. EMI refers to the disturbance that affects the performance of an electronic system due to electromagnetic radiation from an external source. This interference can lead to signal distortion, data loss, or even complete system failure. On the other hand, EMC focuses on the ability of electronic systems to function properly in the presence of electromagnetic interference, ensuring that devices can coexist without adversely affecting one another.

In practical terms, EMI/EMC plays a crucial role in various industries, including telecommunications, automotive, aerospace, and consumer electronics. To ensure the reliable operation of electronic systems, it is necessary to implement effective EMI/EMC measures during the design, testing, and production stages. This involves employing shielding techniques, filtering methods, proper grounding, and careful component placement to minimize electromagnetic interference.

Moreover, compliance with international standards and regulations is essential to guarantee that electronic devices meet the prescribed levels of electromagnetic emissions and immunity. Manufacturers often conduct comprehensive EMI/EMC testing and analysis to assess the performance of their products under different electromagnetic environments and to ensure that they adhere to the necessary industry standards.

EMI/EMC principles and their practical implementation is crucial for maintaining the reliability and performance of electronic systems, enabling the seamless operation of various technologies in today's interconnected world.

Silicon Carbide, Gallium Nitride

Silicon Carbide (SiC) and Gallium Nitride (GaN) are wide-bandgap semiconductors that have gained significant attention in electrical and electronic circuits due to their exceptional properties. They are known for their ability to operate at higher temperatures, voltages, and frequencies compared to traditional silicon-based devices. This makes them ideal for power electronics, including power conversion systems, high-frequency circuits, and radio frequency (RF) devices.

Their incorporation in electrical and electronic circuits is driven by their superior power efficiency, reduced energy losses, and improved thermal management capabilities, which ultimately contribute to the development of more compact, lightweight, and energy-efficient devices. These semiconductors enable the design of smaller and more powerful systems, making them well-suited for various applications, including electric vehicles, renewable energy systems, and aerospace technologies.

However, the compact size of devices employing SiC and GaN can pose challenges related to electromagnetic interference (EMI) and electromagnetic compatibility (EMC). The faster switching speeds and higher clock frequencies associated with these materials can result in increased electromagnetic emissions, leading to potential interference with other electronic components and systems. Therefore, careful consideration of the layout, shielding techniques, and appropriate filtering methods becomes crucial to mitigate the adverse effects of EMI/EMC. Designing effective EMI filters and implementing proper shielding strategies are essential to ensure that the electromagnetic emissions from SiC and GaN-based devices do not interfere with the operation of other nearby electronic circuits, thus maintaining overall system reliability and performance.

Surrogate modelling Techniques

Surrogate modeling, also known as metamodeling, is a technique used in engineering and computational modeling to approximate complex, computationally intensive, or expensive computer simulations with simpler mathematical models. These surrogate models are created using a limited number of simulations or data points, which helps in reducing the computational time and resources required to analyze and optimize the system.

Regarding EMI/EMC, surrogate modeling can be utilized to approximate the behavior of electromagnetic interference (EMI) in electronic systems. By creating simplified mathematical representations of the EMI behavior based on a limited set of data points or simulations, engineers can effectively predict the impact of electromagnetic emissions on the electronic devices. This enables the efficient design of EMI filters and the implementation of appropriate shielding techniques to ensure electromagnetic compatibility (EMC) and minimize the adverse effects of interference on the electronic systems. By using surrogate modeling, engineers can optimize the EMI/EMC design process, leading to the development of more reliable and robust electronic devices.

A. Kriging

Kriging or Gaussian Process modeling is well-known for scalar predictions based on a number of n training samples. Consider a set of n training data, $X = \{x(1), x(2), \dots, x(n)\}^T$, where each $x = \{x_1, x_2, \dots, x_k\}$ with corresponding output $y = \{y(1), y(2), \dots, y(n)\}^T$. Using this training data, we like to predict new responses $\tilde{y}$ given any new input vector $\tilde{x}$. A Kriging meta model provides the prediction based on assumption that the vector formed by the prediction at x , i.e., $\hat{y}(x)$ and the true model responses y , have a joint Gaussian distribution defined by:

$$\begin{pmatrix} \hat{y}(x) \\ y(x) \end{pmatrix} \sim \mathcal{N}_{n+1} \left(\begin{pmatrix} f^T(x)\beta \\ F\beta \end{pmatrix}, \sigma^2 \begin{pmatrix} 1 & r^T(x) \\ r(x) & R \end{pmatrix} \right)$$

where F is the observation (design) matrix of the Kriging metamodel trend given as $F_{ij} = f_j(x(i))$. The vector $r(x)$ describes the cross-correlations between the prediction point x and each of the observations. The matrix R is the correlation matrix with elements. With a Gaussian correlation, the output function predicted for unknown input is given as:

$$\hat{y}(x) = f(x)^T \hat{\beta} + \tilde{r}(x)^T \tilde{R}^{-1} (y - F\hat{\beta})$$

B. Polynomial Chaos Expansion (PCE)

For given n training samples X with corresponding output y , a polynomial approximation of scalar valued function $\hat{y}(x)$ of order m can be written as:

$$\hat{y}(x) = \sum w_i \chi_i(x)$$

where $\chi(x)$ represents multivariate polynomials orthonormal w.r.t the input distribution of the x and w_i indicates corresponding coefficients.

C. Polynomial Chaos-Kriging (PCK)

Kriging interpolates the local variations of a function, whereas PCE approximates the global behavior of Y . Polynomial Chaos-Kriging (PCK) is a method obtained by combining these two techniques to realize a more accurate meta model. The predictor $\hat{y}(x)$ is:

$$\hat{y}(x) = \sum w_i \chi_i(x) + \sigma^2 Z(x, \omega)$$

where first term relates to PCE and second term relates to a stationary Gaussian process.

D. Wideband Kriging (WBK)

Wideband Kriging (WBK) operates, instead of a vector of n normal distributed training data samples as in kriging, with training tuples $\{y_l\}$, $l = 1, \dots, n$ which are vectors of size p with p denoting the number of frequency samples of the data. This is because, in EMC computation, we easily obtain dense information along the frequency axes for a given set of design parameters using frequency-domain simulation. WBK leverages this information to improve scalar kriging. The Gaussian Process on which the broadband Kriging is based is a random matrix $Y \in \mathbb{R}^{p \times n}$ distributed as $Y^T \sim N(\mu 1^T, \sigma^2 \otimes \Psi)$ where $\mu 1^T$ is the mean value matrix, σ^2 the diagonal matrix composed of the variances σ_i^2 , Ψ the correlation matrix and $\otimes$ denotes the Kronecker product. The final broadband predictor of a frequency response is:

$$\hat{\mathbf{y}} = \boldsymbol{\mu} + \boldsymbol{\psi}^T \boldsymbol{\Psi}^{-1} (\mathbf{Y} - \boldsymbol{\mu} \mathbf{1}^T)^T$$

MODEL OF THE EMI FILTER

The filter topology chosen is C-L-C, i.e. there are capacitance's mounted before and after a common mode choke with inductance L_{CM} and parasitic parallel capacitance CL_{CM} . A schematic of the EMI filter connected in CISPR 25 setup for measurement of CM emissions is shown in Fig. 1. The EMI filter is characterized by its Insertion Loss (IL) which is defined as:

$$IL(dB) = 20 \log \left(\frac{U_{CM,w/Filter}}{U_{CM,w/oFilter}} \right)$$

where $U_{CM,w/Filter}$, $U_{CM,w/oFilter}$ denote the CM voltage with filter (as in Fig. 1) and without filter, respectively.

To equip the system with a realistic CM source impedance, we add in series to the voltage source a common mode capacitance (CCM) varying in the same range as that of the CM parasitic capacitance CCM of a DC-DC converter. The voltage $U_{CM,w/oFilter}$ is calculated by removing the filter from this setup. This model is needed to take include the effect of the source impedance.

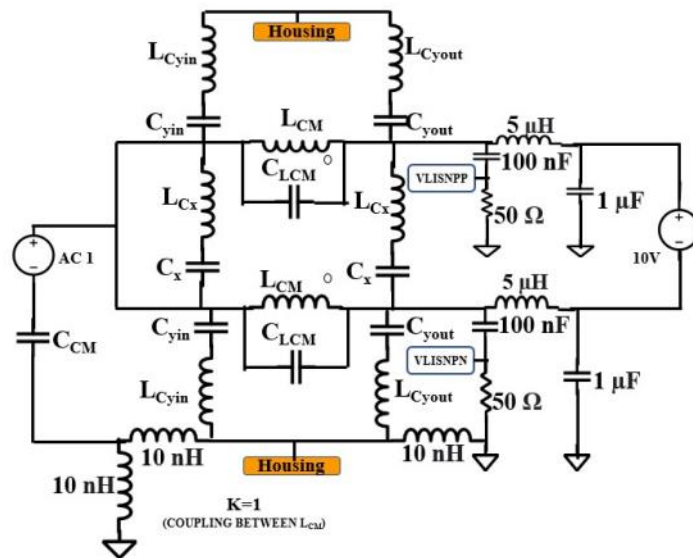


Fig. 1. Schematic of the EMI filter

In total, the IL of the filter has seven design parameters, namely the common mode inductance L_{CM} of the common mode choke, its equivalent parallel capacitance (CL_{CM}), the values of the CY capacitors (C_{yin} and C_{yout}), their parasitic inductance (L_{Cyin} and L_{Cyout}), and the common mode capacitance C_{CM} .

Comparison of Different Surrogate Techniques

A. Building of Surrogate models

The range of filter parameters used for training the surrogate models is as given in Tab. I. UQLAB framework software obtains PCK, PCE, and Kriging(K) default surrogate models. Models are generated for different sizes of input training sets of 250x1, 500x1, and 1000x1 vectors and corresponding output (IL) vectors of length 250x1, 500x1, and 1000x1, respectively. Accordingly, wideband Kriging prediction modelling of the filter uses a training set of length $p = 62$ frequency and $n = 250/500/1000$. MATLAB calls LTSpice to generate the training set required as input for the EMI filter modeling. The Random Latin Hypercube (RLH) sampling method selects training data from the design space mentioned earlier.

TABLE I
RANGES OF THE DESIGN PARAMETERS OF THE EMI FILTER

Name	Range	Function
L_{CM}	1 μ H – 50 μ H	CM choke inductance
C_{CM}	0.2 pF – 200 pF	CM parasitic capacitance
C_{LCM}	0.1 pF – 50 pF	Parallel Capacitance of CM choke inductance
C_{yin}	2.2 nF – 100 nF	Cy capacitance-Input side
L_{Cyin}	4.7 nH–25 nH	Cyin parasitic inductance
C_{yout}	22 nF – 1000 nF	Cy capacitance-Output side
L_{Cyout}	4.7 nH–25 nH	Cyout parasitic inductance

B. Comparison of Models

A comparison of different models w.r.t actual IL is carried out using MATLAB and LTSpice simulation. Fig (2) and Fig (3) show a comparison between the predicted models and the actual LTSpice IL when trained for 500 points and 1000 points, respectively, for design parameter values of $L_{CM} = 20 \mu$ H, $C_{LCM} = 20$ pF, $C_{yin} = 50$ nF, $L_{Cyin} = 10$ nH, $C_{yout} = 500$ nF, $L_{Cyout} = 10$ nH and $C_{CM} = 20$ pF.

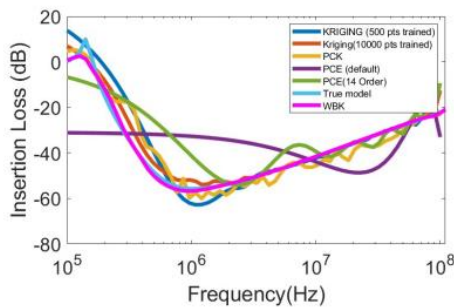


Fig. 2. Typical IL prediction of different models (500 training points)

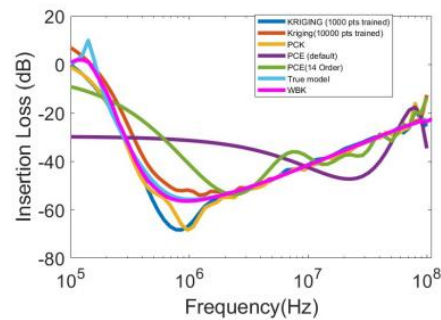


Fig. 3. Typical IL prediction of different models (1000 training points)

C. Quantification of Error

Quantifying the error between the predicted model output and actual IL from LTSpice is done using Normalized Root Mean Square Error (NRMSE). Predicted models built using 500 training points are verified at 10 different testing samples by varying the design parameters using the MATLAB RLH function.

The NRMSE for same is shown in Fig. 4. It can be seen that PCE with default polynomial in UQLAB deviates the most. Detailed comparison between the other models w.r.t training samples and training time is shown in Tab. II. The results indicate that WBK provides the most accurate prediction for a given set of training samples. This accuracy comes at the cost of an excessive training time required to estimate the model hyperparameters at $p = 62$ frequency samples.

It takes 12.7 hours to train Kriging on points equal to the total number of points in WBK, i.e. 15500 (equivalent to 250×62 points in WBK), with a resulting average NRMSE of 0.06, which is comparable to WBK's average NRMSE. Training Kriging on 10,000 points takes 12380 seconds to build a model with an NRMSE of 0.07, which is more than the WBK prediction for 250×62 points. PCK gives a similar average NRMSE of 0.08 with 1000 training points in 107 seconds as in Table II.

The PCK and kriging methods are used from an existing toolbox. WBK is implemented by us in MATLAB and the code can be optimized for speed.

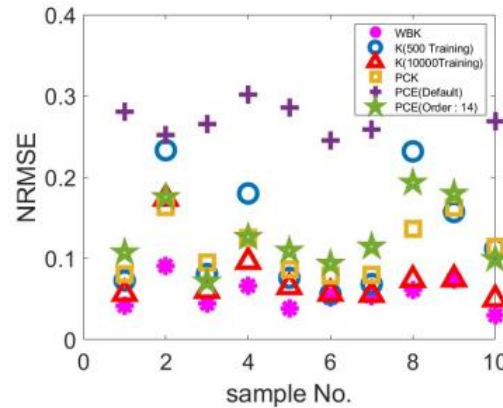


Fig. 4. NRMSE at 10 test samples (Model generated from 500 points)

TABLE II
COMPARISON OF SURROGATE MODELS

Method	Model building time (sec)	Avg. NRMSE (at 10 points)	Training Samples
Kriging	5.37	0.133	250
	18.01	0.1268	500
	59.88	0.0749	1000
	12387	0.076	10000
	45720	0.06	15500
WBK	3600	0.062	250×62 ($n \times p$)
	14400	0.0552	500×62
	36000	0.0472	100×62
PCK	7.06	0.1383	250
	19.16	0.1123	500
	107.9	0.0837	1000

Code Snippets:

```
% clc
% clear all
close all
n = input("Enter the number of samples : ");
X = rlh(n,7,1);
min_Cs = 2.2;
max_Cs = 100;

min_L = 1;
max_L = 50;

min_Csr = 0.2;
max_Csr = 200;

min_Lsr = 4.7;
max_Lsr = 25;

min_EPC = 0.1;
max_EPC = 50;

min_Lsr1 = 4.7;
max_Lsr1 = 25;

min_Cs1 = 22;
max_Cs1 = 1000;

Cs = 10.*((X(:,1).*(log10(max_Cs) - log10(min_Cs)) + log10(min_Cs)));
L = 10.*((X(:,2).*(log10(max_L) - log10(min_L)) + log10(min_L)));
Csr = 10.*((X(:,3).*(log10(max_Csr) - log10(min_Csr)) + log10(min_Csr)));
Lsr = 10.*((X(:,4).*(log10(max_Lsr) - log10(min_Lsr)) + log10(min_Lsr)));
EPC = 10.*((X(:,5).*(log10(max_EPC) - log10(min_EPC)) + log10(min_EPC)));
Lsr1 = 10.*((X(:,6).*(log10(max_Lsr1) - log10(min_Lsr1)) + log10(min_Lsr1)));
Cs1 = 10.*((X(:,7).*(log10(max_Cs1) - log10(min_Cs1)) + log10(min_Cs1)));

freq_par = logspace(log10(1e5),log10(1e8),n);
freq_par = freq_par(randperm(length(freq_par)));
freq_par = transpose(freq_par);

%% Simulation using LTSpice to get the required values

% tic
% time2 = 0.5e-3 : Ts : 1.2e-3;
dos('XVIIx64 -netlist filternewmodelv.asc')
pause(1)
dos('XVIIx64 -b filternewmodelv.net')
pause(1)
y = LTSpice2Matlab('filternewmodelv.raw',[1]) ;

for i = 1 : n
    clear temp
    % Writing the required variables into LTSpice using MATLAB
    str = ".param Cs="+num2str(Cs(i))+ "n" + " L="+num2str(L(i))+ "u" +
Csr="+num2str(Csr(i))+ "p" + " Lsr="+num2str(Lsr(i))+ "n" + " EPC="+num2str(EPC(i))+ "p"
+ " Cs1="+num2str(Cs1(i))+ "n" + " Lsr1="+num2str(Lsr1(i))+ "n";
    fileid = fopen('D:\Study\IISC\freq_domain_LTSpice\7D\EPCnew.txt','w');
    fprintf(fileid,'%s',str);
```

```

str2 = ".ac list="+num2str(freq_par(i));
fileid = fopen('D:\Study\IISC\freq_domain_LTspice\7D\ac.txt','w');
fprintf(fileid,'%s',str2);

% Generating LTSpice netlist using MATLAB
dos('XVIIx64 -netlist filternewmodelv.asc')
pause(1)
% Running the netlist
dos('XVIIx64 -b filternewmodelv.net')
pause(1)

% Processing of raw data
y = LTspice2Matlab('filternewmodelv.raw',[12 13 17 19]);
temp(:,1) = y.variable_mat(1,:);
temp(:,2) = y.variable_mat(2,:);
temp(:,3) = y.variable_mat(3,:);
temp(:,4) = y.variable_mat(4,:);

output(:,i) = abs((temp(:,1)+temp(:,2))/(temp(:,3)+temp(:,4)));
output_log(:,i) = 20*log10(output(:,i));

end
%%
output_log = transpose(output_log);
%% Metamodels 1.Kriging
uqlab
InputOpts.Marginals(1).Type = 'Uniform';
InputOpts.Marginals(1).Parameters = [2.2, 100];
InputOpts.Marginals(2).Type = 'Uniform';
InputOpts.Marginals(2).Parameters = [1, 50];
InputOpts.Marginals(3).Type = 'Uniform';
InputOpts.Marginals(3).Parameters = [0.2, 200];
InputOpts.Marginals(4).Type = 'Uniform';
InputOpts.Marginals(4).Parameters = [4.7, 25];
InputOpts.Marginals(5).Type = 'Uniform';
InputOpts.Marginals(5).Parameters = [0.1, 50];
InputOpts.Marginals(6).Type = 'Uniform';
InputOpts.Marginals(6).Parameters = [22, 1000];
InputOpts.Marginals(7).Type = 'Uniform';
InputOpts.Marginals(7).Parameters = [4.7, 25];
InputOpts.Marginals(8).Type = 'Uniform';
InputOpts.Marginals(8).Parameters = [1e5, 1e8];

inputs = [Cs,L,Csr,Lsr,EPC,Cs1,Lsr1,freq_par];
X_train = inputs;
myInput = uq_createInput(InputOpts);
ModelOpts.mHandle = @(X_train) uq_kriging_predict(X_train, X_training,
Y_training);
myModel = uq_createModel(ModelOpts);

MetaOpts.Type = 'Metamodel';
MetaOpts.MetaType = 'Kriging';
MetaOpts.Input = myInput;
MetaOpts.FullModel = myModel;
MetaOpts.ExpDesign.X = X_train;
MetaOpts.ExpDesign.Y = output_log;
myKriging = uq_createModel(MetaOpts);
uq_print(myKriging)

```

```

%% PCE MODEL
% uqlab
output_log = transpose(output_log);
Inputopts.Marginals(1).Type = 'Uniform';
Inputopts.Marginals(1).Parameters = [2.2, 100];
Inputopts.Marginals(2).Type = 'Uniform';
Inputopts.Marginals(2).Parameters = [1, 50];
Inputopts.Marginals(3).Type = 'Uniform';
Inputopts.Marginals(3).Parameters = [0.2, 200];
Inputopts.Marginals(4).Type = 'Uniform';
Inputopts.Marginals(4).Parameters = [4.7, 25];
Inputopts.Marginals(5).Type = 'Uniform';
Inputopts.Marginals(5).Parameters = [0.1, 50];
Inputopts.Marginals(6).Type = 'Uniform';
Inputopts.Marginals(6).Parameters = [22, 1000];
Inputopts.Marginals(7).Type = 'Uniform';
Inputopts.Marginals(7).Parameters = [4.7, 25];
Inputopts.Marginals(8).Type = 'Uniform';
Inputopts.Marginals(8).Parameters = [1e5, 1e8];

inputs = myKriging.ExpDesign.X;
X_train_pce = inputs;
myInput_pce = uq_createInput(Inputopts);
Modelopts.mHandle = @(X_train_pce) uq_pce_predict(X_train_pce, X_training,
Y_training);
myModel_pce = uq_createModel(Modelopts);

Metaopts.Type = 'Metamodel';
Metaopts.MetaType = 'PCE';
Metaopts.Input = myInput_pce;
Metaopts.FullModel = myModel_pce;
Metaopts.ExpDesign.X = X_train_pce;
Metaopts.ExpDesign.Y = output_log;
myPCE = uq_createModel(Metaopts);
uq_print(myPCE)

%% PCK MODEL
output_log = transpose(output_log);
Inputopts.Marginals(1).Type = 'Uniform';
Inputopts.Marginals(1).Parameters = [2.2, 100];
Inputopts.Marginals(2).Type = 'Uniform';
Inputopts.Marginals(2).Parameters = [1, 50];
Inputopts.Marginals(3).Type = 'Uniform';
Inputopts.Marginals(3).Parameters = [0.2, 200];
Inputopts.Marginals(4).Type = 'Uniform';
Inputopts.Marginals(4).Parameters = [4.7, 25];
Inputopts.Marginals(5).Type = 'Uniform';
Inputopts.Marginals(5).Parameters = [0.1, 50];
Inputopts.Marginals(6).Type = 'Uniform';
Inputopts.Marginals(6).Parameters = [22, 1000];
Inputopts.Marginals(7).Type = 'Uniform';
Inputopts.Marginals(7).Parameters = [4.7, 25];
Inputopts.Marginals(8).Type = 'Uniform';
Inputopts.Marginals(8).Parameters = [1e5, 1e8];

% inputs = [Cs,L,Csr,Lsr,EPC,Cs1,Lsr1,freq_par];
X_train = myKriging.ExpDesign.X;
myInput_pck = uq_createInput(Inputopts);
Modelopts.mHandle = @(X_train)uq_pck_predict(X_train, X_training, Y_training);

```

```

myModel_pck = uq_createModel(Modelopts);

Metaopts.Type = 'Metamodel';
Metaopts.MetaType = 'PCK';
Metaopts.Input = myInput_pck;
Metaopts.FullModel = myModel_pck;
Metaopts.ExpDesign.X = X_train;
Metaopts.ExpDesign.Y = output_log;
myPCK = uq_createModel(Metaopts);
uq_print(myPCK)

%% PCK modified_1.Exp 2.
% output_log = transpose(output_log);
output_log = transpose(output_log);
Inputopts.Marginals(1).Type = 'Uniform';
Inputopts.Marginals(1).Parameters = [2.2, 100];
Inputopts.Marginals(2).Type = 'Uniform';
Inputopts.Marginals(2).Parameters = [1, 50];
Inputopts.Marginals(3).Type = 'Uniform';
Inputopts.Marginals(3).Parameters = [0.2, 200];
Inputopts.Marginals(4).Type = 'Uniform';
Inputopts.Marginals(4).Parameters = [4.7, 25];
Inputopts.Marginals(5).Type = 'Uniform';
Inputopts.Marginals(5).Parameters = [0.1, 50];
Inputopts.Marginals(6).Type = 'Uniform';
Inputopts.Marginals(6).Parameters = [22, 1000];
Inputopts.Marginals(7).Type = 'Uniform';
Inputopts.Marginals(7).Parameters = [4.7, 25];
Inputopts.Marginals(8).Type = 'Uniform';
Inputopts.Marginals(8).Parameters = [1e5, 1e8];

% inputs = [Cs,L,Csr,Lsr,EPC,Cs1,Lsr1,freq_par];
X_train = training_full;
myInput_matern = uq_createInput(Inputopts);
Modelopts.mHandle = @(X_train)uq_pck_predict(X_train, X_training, Y_training);
myModel_matern = uq_createModel(Modelopts);

Metaopts.Type = 'Metamodel';
Metaopts.MetaType = 'PCK';
Metaopts.Kriging.Corr.Type = 'separable';
Metaopts.Kriging.Corr.Family = 'matern-5_2';
Metaopts.Input = myInput_matern;
Metaopts.FullModel = myModel_matern;
Metaopts.ExpDesign.X = X_train;
Metaopts.ExpDesign.Y = output_log;
myPCK_matern = uq_createModel(Metaopts);
uq_print(myPCK_matern);

%% Testing for some 100 samples , keeping all param constant except freq, freq to
be varied to observe a plot n compare with LTspice result :)
% 1st-: Cs = 10 , L = 25 , Csr = 20 , Lsr = 10 , EPC = 10 , Cs1 = 100 ,
% Lsr1 = 10 , freq = logspace , 1e5, 1e8 , 100

testing_Cs = 20*ones(100,1);
testing_L = 15*ones(100,1);
testing_Csr = 40*ones(100,1);
testing_Lsr = 15*ones(100,1);
testing_EPC = 20*ones(100,1);

```



```

testing_Cs1 = 200*ones(100,1);
testing_Lsr1 = 15*ones(100,1);
testing_freq = logspace(log10(1e5),log10(1e8),100);
testing_freq = transpose(testing_freq);
testing_inputs =
[testing_Cs,testing_L,testing_Csr,testing_Lsr,testing_EPC,testing_Cs1,testing_Lsr1
,testing_freq];
% tic
% result_krg = uq_evalModel(myKriging,testing_inputs);
% toc
%%
dos('XVIIx64 -netlist filternewmodelv.asc')
pause(1)
dos('XVIIx64 -b filternewmodelv.net')
pause(1)
y = LTspice2Matlab('filternewmodelv.raw',[1]) ;

for i = 1 : 100
    % clear temp
    % Writing the required variables into LTSpice using MATLAB
    str = ".param Cs="+num2str(testing_Cs(i))+ "n"+"
L="+num2str(testing_L(i))+ "u"+" Csr="+num2str(testing_Csr(i))+ "p"+"
Lsr="+num2str(testing_Lsr(i))+ "n"+" EPC="+num2str(testing_EPC(i))+ "p" +"
Cs1="+num2str(testing_Cs1(i))+ "n" +" Lsr1="+num2str(testing_Lsr1(i))+ "n";
    fileid = fopen('D:\Study\IISC\freq_domain_LTspice\7D\EPCnew.txt','w');
    fprintf(fileid,'%s',str);

    str2 = ".ac list="+num2str(testing_freq(i));
    fileid = fopen('D:\Study\IISC\freq_domain_LTspice\7D\ac.txt','w');
    fprintf(fileid,'%s',str2);

    % Generating LTSpice netlist using MATLAB
    dos('XVIIx64 -netlist filternewmodelv.asc')
    pause(1)
    % Running the netlist
    dos('XVIIx64 -b filternewmodelv.net')
    pause(1)

    % Processing of raw data
    y = LTspice2Matlab('filternewmodelv.raw',[12 13 17 19]);
    temp_test(:,1) = y.variable_mat(1,:);
    temp_test(:,2) = y.variable_mat(2,:);
    temp_test(:,3) = y.variable_mat(3,:);
    temp_test(:,4) = y.variable_mat(4,:);

    output_test(:,i) =
abs((temp_test(:,1)+temp_test(:,2))/(temp_test(:,3)+temp_test(:,4)));
    output_log_test(:,i) = 20*log10(output_test(:,i));

end

%%
tic
result_krg = uq_evalModel(myKriging,testing_inputs);
toc
tic
result_pck = uq_evalModel(myPCK,testing_inputs);
toc

```

```

% result_pce = uq_evalModel(myPCE,testing_inputs);
tic
result_pck_linear = uq_evalModel(myPCK_Linear,testing_inputs);
toc
tic
result_pck_exponential = uq_evalModel(myPCK_modified,testing_inputs);
toc
tic
result_pck_matern = uq_evalModel(myPCK_matern,testing_inputs);
toc

%% plotting
uq_figure
semilogx(testing_freq,result_pck,LineWidth=1.5);
hold on;
semilogx(testing_freq,result_pck_linear,LineWidth=1.5);
hold on;
semilogx(testing_freq ,result_pck_exponential,LineWidth=1.5);
hold on;
semilogx(testing_freq,result_pck_matern,LineWidth=1.5);
hold on;
semilogx(testing_freq,output_log_test,LineWidth=2.5);
legend('normal PCK','PCK linear','PCK exponential','PCK matern','True model');
xlabel('Frequency');
ylabel('Insertion Loss (dB)');
title('Comparison of models : 500 training samples , 100 testing')
%%
% training_full = myKriging.ExpDesign.X;
% semilogy(1:500,sort(freq_par),'o');

```

Conclusion

This investigation provides a concise assessment of the practicality of various surrogate modeling techniques in accurately and efficiently simulating the Insertion Loss behavior of the EMI filter.

It also underscores the significance of the quantity of training data points, emphasizing how this factor impacts both the training duration and the precision of the models. The findings demonstrate that while Wideband Kriging (WBK) yields the most precise predictions, it necessitates a substantial investment of time during the training phase.

Conversely, for scenarios where an exceptionally precise forecast is not imperative, Kriging and Polynomial Chaos Kriging (PCK) methods emerge as viable alternatives, exhibiting shorter training times without compromising the overall effectiveness of the modeling process.

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